

SSIE-500: Homework 5

Measuring Complexity of Languages (1)

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Deadline: November 10, 2025 (1:30pm Eastern Time)

1. Complete the following tasks for three different human languages of your choice (e.g., English, Spanish, French, Chinese, Arabic, Hindi, Korean, Japanese, etc.). You can use any programming language for computation.
 - (a) Read in a sufficiently long text file written in that language from online sources (e.g., a copyright-free novel).
 - (b) Implement the calculation of *information entropy* of the frequency distribution of symbols that appear in the text. For languages that use the Latin alphabet, you can keep their cases as is. You can keep spaces, punctuations, and other symbols in this calculation too.
 - (c) Implement the calculation of *mutual information* that exists between two consecutive symbols in the text.
2. Compare the results obtained for the three human languages and discuss what you have learned from the results for those languages.
3. Produce a PDF file of your report using $\text{\LaTeX}$, properly displaying your program code, the results, and any comments/discussions. Submit it to the Assignment page on Brightspace by the deadline.